TITANIA Results: Automated Data Error Cleaning Impact on Federated Learning Utility and Fairness

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I. Datasets and Models

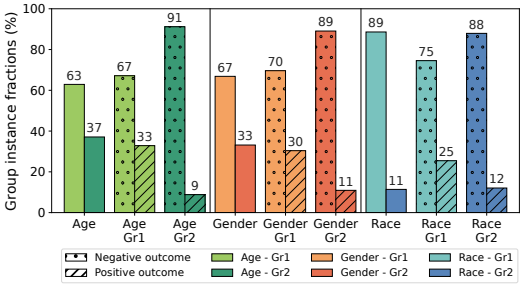
TITANIA includes several well-known tabular datasets, as listed in Table I, that contain sensitive attributes, such as gender, race, and age. Adult and KDD contain census data and are used to predict whether an individual's annual income is high or low [1], [2]. ARS is a sensor-based dataset that contains human activities of elderly participants and aims to determine whether participants are lying or not [3]. Heart is a healthcare dataset containing patient measurements used to predict the presence of heart diseases [4]. MEPS is also a medical dataset that contains in-person data access to hospitals, and its purpose is to predict whether a person has used medical facilities a lot of times [5]. Furthermore, TITANIA includes three classification models to predict data outcomes: Logistic Regression (LR) [6], Support Vector Machine (SVM) [7], and Multi-Layer Perceptron (MLP) [8].

TABLE I
DATASET CHARACTERISTICS

Dataset	Topic	#Records	#Attr.	Sensitive attr.	Label
KDD	Finance	299,285	41	*gender, race, age*	High/low income
ARS	Healthcare	75,128	9	*gender*	Lying/non-lying position
Heart	Healthcare	70,000	11	*gender, age*	Heart disease or not
Adult	Finance	48,842	15	*gender, race, age*	High/low income
MEPS	Healthcare	33,400	42	*gender, race*	Visited few times/a lot of times

In Figure 1, we further analyze the demographic groups in the Adult dataset. We observe disparities between demographic groups, in particular, regarding the proportion of positive and negative outcomes. Here, the datas contain 67% of men *vs*. 33% of women, and there are almost three times more men than women who have positive outcomes (*i.e*., have a high salary). These disparities also appear in the other datasets, as shown in Figure 2.

Fig. 1. Demographic disparities in Adult



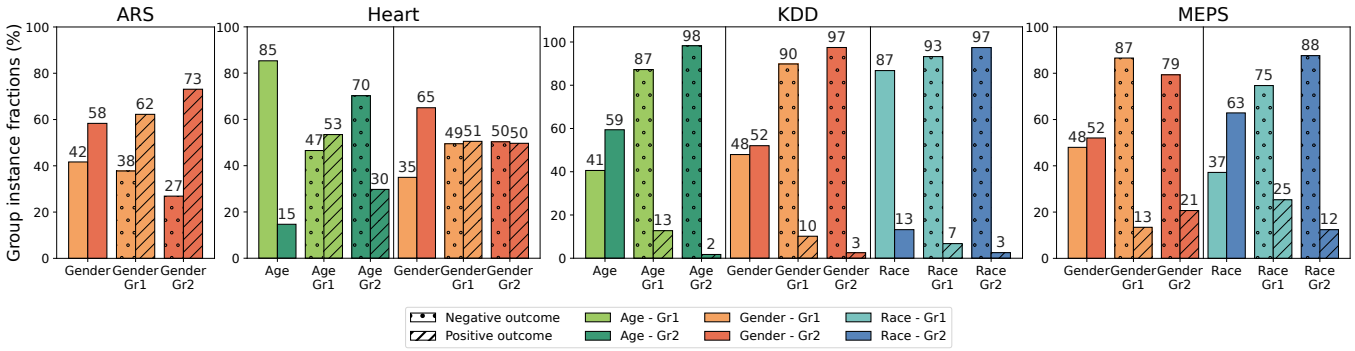
II. Data Cleaning Methods

The TITANIA framework includes several data cleaning methods that fall into one of the following three categories depending on the type of corrupted data covered by the method: (i) missing values, (ii) outliers, or (iii) label errors, as detailed below.

Missing Value Cleaning Methods. Missing values are flagged during the data loading stage, and their imputation is performed in different ways [9]. For instance, if a missing value is a numerical attribute, the imputed value can be the most frequently occurring value (*i.e*., the mode) in the set of values of that attribute. The imputed missing numerical value can also be the mean of the set of values for the attribute in question. In the case of a categorical missing value, data imputation is based on the mode value of the attribute. In an FL system, the calculation of mode or mean values is performed either independently by each FL client based on its local data, or based on the global statistical information of all FL clients' data. In the latter case, the global mean or mode values are computed using a distributed protocol between the FL clients and server. In TITANIA, we implement missing value cleaning methods denoted as *mv-mn-l*, *mv-mn-g*, *mv-md-l*, and *mv-md-g*, where *mv* refers to missing value cleaning, and *mn*, *md*, *l*, and *g* represent the mean, mode, local, and global, respectively. Recent solutions also propose personalized FL approaches specifically targeting heterogeneous data imputation, such as the Cafe system [10] that we integrate into TITANIA and is referred to as *mv-cf*.

Outlier Cleaning Methods. Handling data outliers is another classical technique for cleaning the numerical data [9]. To detect outliers, it is necessary to determine the interval of values allowed for an attribute. For instance, the interquartile range (IQR) can be used to determine the interval of allowed values related to the 25th and 75th percentiles of the considered attributes, denoted p_{25} and p_{75}. Thus, with $iqr = p_{75} - p_{25}$, values outside the interval $[p_{25} - \gamma \cdot iqr, p_{75} + \gamma \cdot iqr]$ are considered outliers, with $\gamma \in \mathbb{R}^+$. Another method to determine the interval of allowed values for an attribute is through the standard deviation (STD) and mean value of that attribute. Values outside the interval $[mean - \mu \cdot std, mean + \mu \cdot std]$ are considered outliers, with $\mu \in \mathbb{R}^+$. Once an interval of allowed values is defined, there are several ways to handle outliers. For instance, an extreme approach is to exclude from the dataset the samples that contain outliers, but this has the

Fig. 2. Demographic disparities in different datasets



risk of excluding too much data. Other techniques induce the replacement of outliers with median, mean, or mode values of the underlying attribute. Furthermore, as mentioned earlier, in an FL system, the calculation of data statistical values is performed either independently by each FL client based on its local data, or based on the global statistical information of all FL clients' data. Thus, TITANIA implements 12 outlier cleaning methods, denoted as *ol-iqr-mn-l*, *ol-iqr-mn-g*, *ol-iqr-md-l*, *ol-iqr-md-g*, *ol-iqr-dl-l*, *ol-iqr-dl-g*, *ol-std-mn-l*, *ol-std-mn-g*, *ol-std-md-l*, *ol-std-md-g*, *ol-std-dl-l*, and *ol-std-dl-g*, where *ol* refers to outlier cleaning, *iqr* and *std* correspond to IQR- and STD-based outlier detection techniques, *mn*, *md* and *dl* represent the mean, mode or deletion approaches, *l* and *g* represent the local and global, respectively.

Label Error Cleaning Methods. Data label errors are frequently introduced by human errors, particularly when the annotation process is tedious. Existing techniques to detect corrupt labels follow the same general approach, which consists of training a model on the data to learn noisy *vs.* correct labels [11]–[13]. For instance, a logistic regression model can be used to detect corrupt labels [11]. Thus, in a dataset with a binary outcome (*cf.* §I), correcting corrupt labels involves flipping their values. TITANIA implements this method for label cleaning, denoted as *le-cl*, as well as its variant *le-cl-dl* where samples are deleted if their labels are corrupted. In addition, recent works propose FL approaches specifically targeting heterogeneous data labels, such as the FedCorr system [14] that we integrate into TITANIA and is referred to as *le-fc*, as well as its variant *le-fc-dl* that deletes samples with label errors.

III. EXPERIMENTAL SETUP

The TITANIA framework is implemented using Python 3.13.5. Five datasets and three models presented in §I are considered in our experiments. As described in §II, TITANIA includes 12 outlier cleaning methods, 5 missing value cleaning methods (among which is the Cafe library [10]), and 4 label error cleaning methods (*i.e.*, the Cleanlab library [15], and the FedCorr library [14]). Outlier cleaning methods based on IQR have $\gamma = 1.5$, and those based on STD have $\mu = 3$ [16], [17]. We also consider a baseline consisting of raw unclean datasets for which the only modification was to remove records with missing values in KDD and Adult datasets. Handling missing values in the baseline is mandatory to allow model training. Sample deletion is frequently used when the dataset is large, or the percentage of deleted samples is small, and when the missingness is unrelated to output classes or to any other features in the dataset (which is the case for KDD and Adult) [18]. Furthermore, unless otherwise stated, all FL experiments are run with ten clients. Due to space limitation, we report on only results of globally computed FL metrics. The per client results can be found in [19]. To account for randomness, each experiment is repeated five times, and each time, all types of metrics are measured at each of the last 20 FL rounds, resulting in 100 paired measured metric for each t-test. All experiments are run on machines equipped with two Intel Xeon Gold 6130 CPUs (16 cores each) and 192 GB of RAM. In our empirical study, we collect extensive FL traces consisting of 2,365 workloads for 1,764,750 trace records. The TITANIA framework is publicly available, along with the collected FL workload traces and detailed hyper-parameters (*cf.* Table II) for reproducibility. All materials are available at [19].

TABLE II
HYPERPARAMETERS. EACH MODEL'S HYPERPARAMETERS WERE OPTIMIZED FOR THE UNDERLYING DATASET. BATCH SIZE IS 128 WITH ALL MODELS, LR AND MLP USE THE BINARY CROSS-ENTROPY LOSS FUNCTION, AND SVM USES THE BINARY HINGE LOSS FUNCTION

Dataset	Model	Optimizer	Learning rate
Adult	LR	AdamW	0.0001
	MLP (64, 32)	RMSprop	0.0001
	SVM	Adam	0.001
ARS	LR	RMSprop	0.01
	MLP (64, 32)	Adagrad	0.01
	SVM	RMSprop	0.01
Heart	LR	RMSprop	0.001
	MLP (64, 32)	Adagrad	0.01
	SVM	RMSprop	0.03
KDD	LR	SGD	0.01
	MLP (64, 32)	SGD	0.003
	SVM	SGD	0.01
MEPS	LR	Adagrad	0.001
	MLP (64, 32)	Adagrad	0.001
	SVM	RMSprop	0.0003

IV. Overall Impact of Data Cleaning

TABLE III
IMPACT OF DATA CLEANING ON UTILITY AND FAIRNESS METRICS

Utility	worse	insig.	better
Accuracy	49%	11%	**40%**
Recall	61%	12%	27%
Precision	58%	12%	30%
F1-score	**62%**	10%	28%

Fairness	worse	insig.	better
AOD	47%	6%	47%
DcI	48%	7%	45%
EOD	47%	10%	43%
DI	**52%**	6%	42%
SPD	41%	6 %	**53%**

TABLE IV
DETAILED IMPACT OF DATA CLEANING ON DIFFERENT FAIRNESS METRICS

Fairness	worse	insig.	better
AOD_{gender}	50%	9%	41%
AOD_{age}	46%	2%	52%
AOD_{race}	44%	4%	52%
DcI_{gender}	50%	9%	41%
DcI_{age}	60%	3%	37%
DcI_{race}	34%	8%	57%
EOD_{gender}	50%	15%	35%
EOD_{age}	45%	6%	49%
EOD_{race}	44%	7%	49%
DI_{gender}	56%	8%	36%
DI_{age}	57%	4%	39%
DI_{race}	38%	6%	56%
SPD_{gender}	50%	10%	40%
SPD_{age}	38%	2%	60%
SPD_{race}	30%	4%	66%

V. Deep Dive into Data Cleaning Methods

TABLE V
IMPACT OF DIFFERENT DATA CLEANING METHODS ON FL MODEL FAIRNESS AND UTILITY

OUTLIER CLEANING METHODS

Outlier cleaning method	Utility			Fairness		
	worse	insig.	better	worse	insig.	better
ol-iqr-md-l	**77%**	10%	13%	57%	5%	38%
ol-iqr-md-g	**77%**	13%	10%	58%	4%	38%
ol-iqr-mn-l	68%	12%	20%	**66%**	6%	28%
ol-iqr-mn-g	65%	13%	22%	**66%**	7%	27%
ol-iqr-dl-l	76%	2%	22%	41%	1%	**58%**
ol-iqr-dl-g	76%	2%	22%	41%	2%	57%
ol-std-md-l	55%	23%	22%	47%	18%	35%
ol-std-md-g	53%	18%	**29%**	48%	11%	41%
ol-std-mn-l	42%	32%	27%	46%	16%	38%
ol-std-mn-g	45%	28%	27%	48%	14%	38%
ol-std-dl-l	65%	20%	15%	38%	6%	56%
ol-std-dl-g	67%	18%	15%	38%	5%	57%

MISSING VALUE CLEANING METHODS

Missing value cleaning method	Utility			Fairness		
	worse	insig.	better	worse	insig.	better
mv-md-l	63%	4%	33%	32%	10%	**58%**
mv-md-g	58%	0%	**42%**	45%	2%	53%
mv-mn-l	58%	0%	**42%**	38%	9%	53%
mv-mn-g	58%	0%	**42%**	45%	2%	53%
mv-cf	**79%**	4%	17%	**57%**	4%	39%

LABEL ERROR CLEANING METHODS

Label error cleaning method	Utility			Fairness		
	worse	insig.	better	worse	insig.	better
le-cl	0%	3%	**97%**	**49%**	8%	43%
le-cl-dl	0%	5%	95%	44%	7%	49%
le-fc	**72%**	0%	28%	39%	1%	**60%**
le-fc-dl	**72%**	0%	28%	39%	3%	58%

VI. Impact of Data Cleaning on Different Datasets

TABLE VI
IMPACT OF DATA CLEANING ON DIFFERENT DATASETS

Dataset	Utility			Fairness			Detected errors		
	worse	insig.	better	worse	insig.	better	outliers	miss. val.	lab. err.
Adult	**80%**	2%	18%	**55%**	5%	40%	✓	✓	✓
KDD	68%	1%	31%	38%	4%	**58%**	✓	✓	✓
Heart	56%	9%	35%	46%	5%	49%	✓	✗	✓
ARS	29%	48%	23%	43%	31%	26%	✓	✗	✓
MEPS	43%	4%	**53%**	53%	7%	40%	✓	✗	✓

TABLE VII
ACCURACY OF THE LR BASELINE FOR EACH DATASET

Dataset	Accuracy
Adult	85.0%
KDD	95.4%
Heart	72.2%
ARS	99.7%
MEPS	83.9%

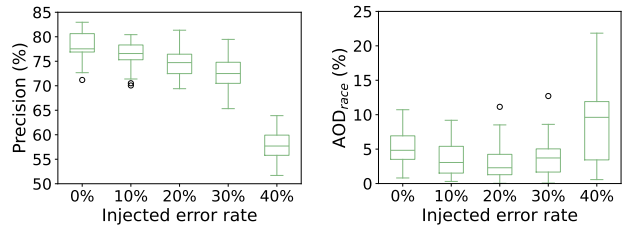
VII. Joint Impact on Utility and Fairness

TABLE VIII
JOINT IMPACT OF DATA CLEANING ON FAIRNESS AND UTILITY

		Utility		
		worse	insig.	better
Fairness	worse	31%	2%	14%
	insig.	3%	2%	2%
	better	30%	2%	14%

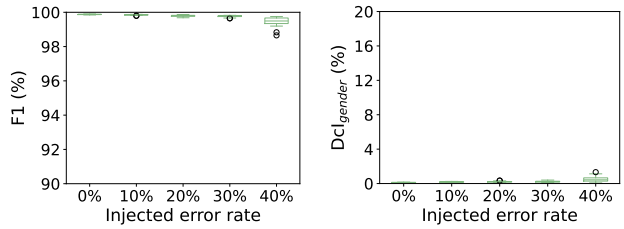
VIII. Data Cleaning Under Various Error Rates

Fig. 3. Impact of data cleaning with various rates of error injection – Adult/LR



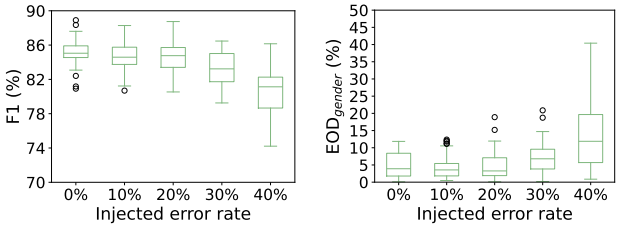
(a) Impact on utility (b) Impact on fairness

Fig. 7. Impact of data cleaning with various rates of error injection ARS/LR



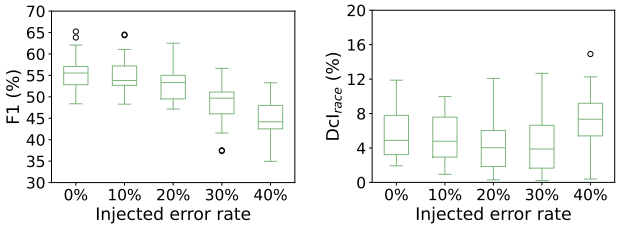
(a) Impact on utility (b) Impact on fairness

Fig. 4. Impact of data cleaning with various rates of error injection Heart/LR



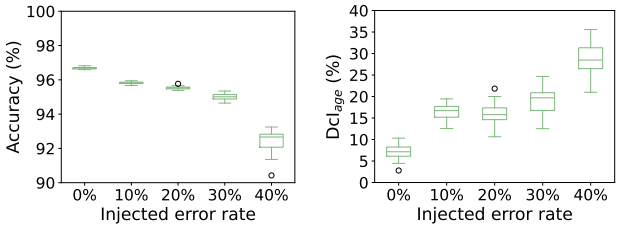
(a) Impact on utility (b) Impact on fairness

Fig. 5. Impact of data cleaning with various rates of error injection MEPS/LR



(a) Impact on utility (b) Impact on fairness

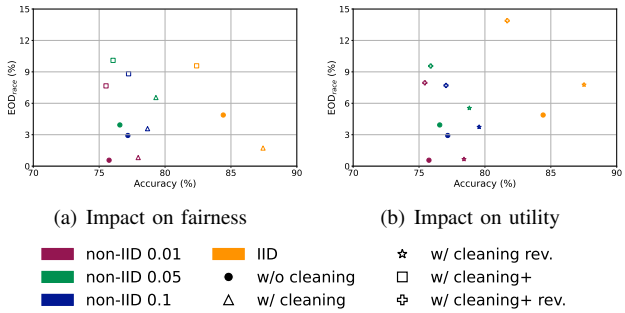
Fig. 6. Impact of data cleaning with various rates of error injection KDD/LR



(a) Impact on utility (b) Impact on fairness

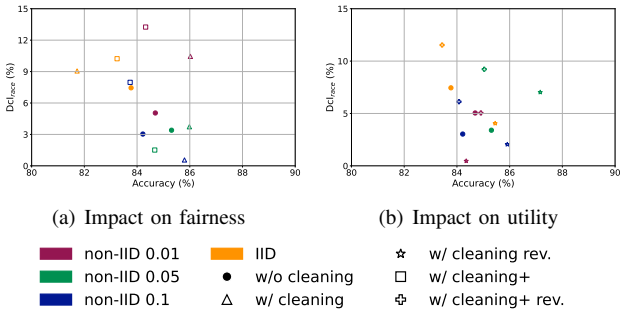
IX. Data Cleaning Under Various FL Non-IID Settings

Fig. 8. Data cleaning under various FL non-IID settings – Adult/LR



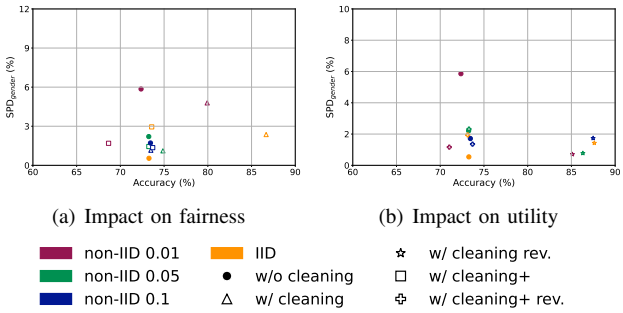
(a) Impact on fairness (b) Impact on utility

Fig. 9. Data cleaning under various FL non-IID settings – MEPS/LR



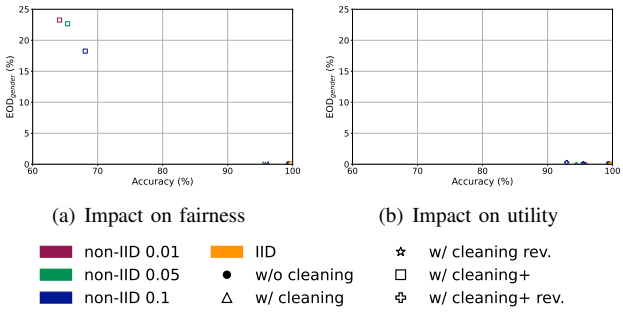
(a) Impact on fairness (b) Impact on utility

Fig. 10. Data cleaning under various FL non-IID settings – Heart/LR



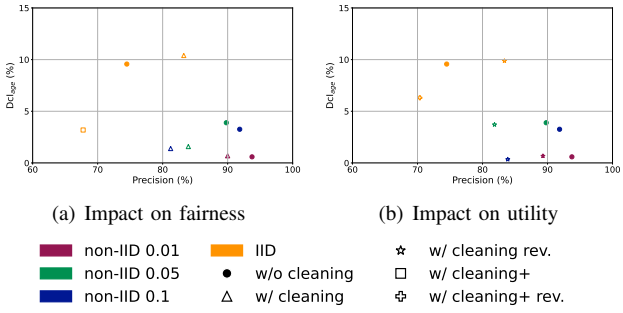
(a) Impact on fairness (b) Impact on utility

Fig. 11. Data cleaning under various FL non-IID settings – ARS/LR



(a) Impact on fairness (b) Impact on utility

Fig. 12. Data cleaning under various FL non-IID settings – KDD/LR



(a) Impact on fairness (b) Impact on utility

X. Data Cleaning *vs.* Bias Mitigation

Fig. 13. Combining data cleaning with bias mitigation – Adult/LR

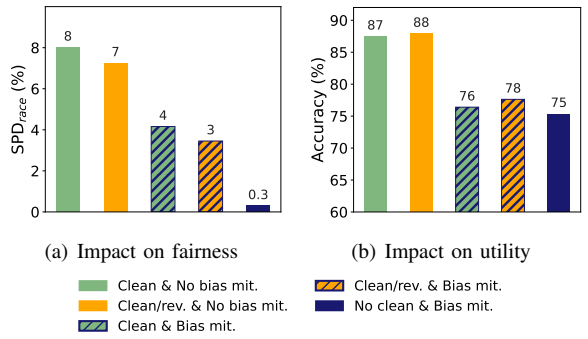


Fig. 14. Combining data cleaning with bias mitigation Heart/LR

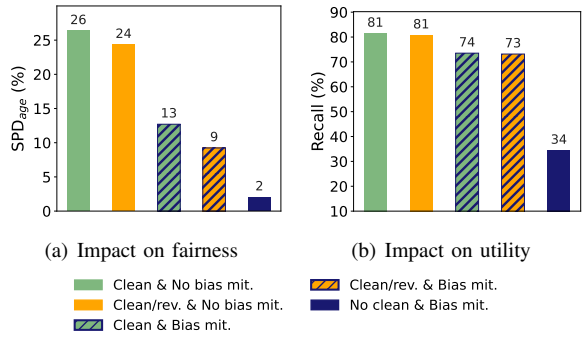


Fig. 15. Combining data cleaning with bias mitigation ARS/LR

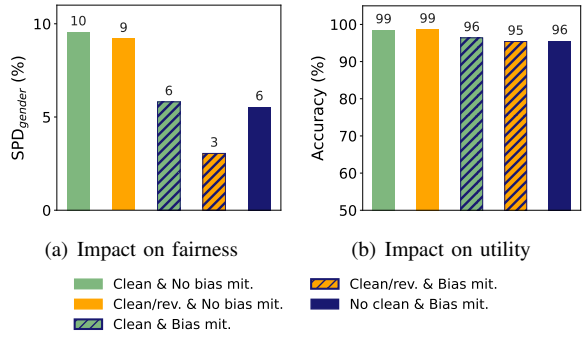


Fig. 17. Combining data cleaning with bias mitigation KDD/LR

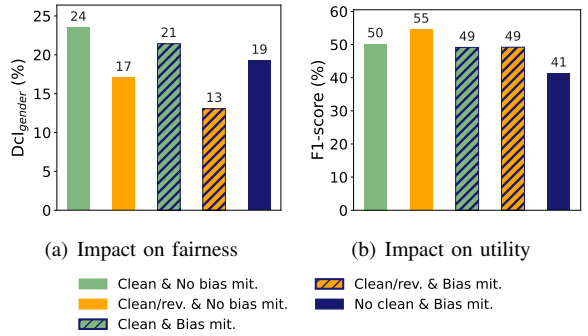
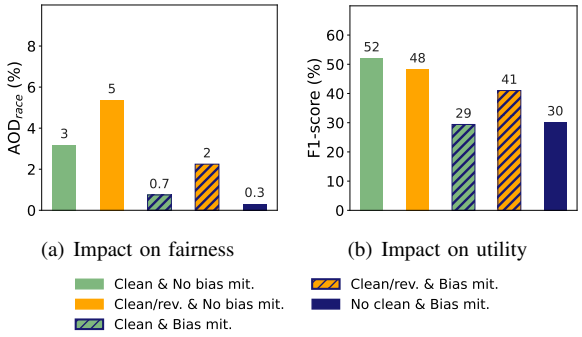


Fig. 16. Combining data cleaning with bias mitigation MEPS/LR



XI. Ranking Cleaning Methods Based on Constraint Metrics

Titania's experimental traces can also serve to guide the best choice of cleaning method for maximizing utility-fairness trade-off. Since evaluation metrics are generally specific to use cases, we provide a ranking tool based on constraint metrics to advise users on the cleaning methods best suited to their specific applications. For this purpose, users need to specify one or several metrics (whether utility or fairness) to generate a personalized report that ranks the different cleaning methods for each scenario of this study (datasets and ML models). Before ranking, note that the evaluation metrics (average among runs) of each cleaning method are discretized into intervals of size w to group the methods with similar metrics. Table IX illustrates a ranking example for each dataset (only scenarios using the logistic regression model). F1-score is chosen as the first comparison metric, DcI$_{gender}$ is used as a second metric to break ties, and w is set to 1% for the discretization of these metrics.

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TABLE IX
DATA CLEANING METHODS RANKING, WITH CONSTRAINTS ON F1-SCORE AND DcI_{gender}

Adult	KDD	Heart	MEPS	ARS
1. *no-cleaning*	1. *no-cleaning*	1. *ol-std-mn-g*	1. *ol-std-mn-l*	1. *ol-std-mn-g*
2. *ol-std-md-g*	2. *ol-iqr-mn-g*	1. *ol-std-mn-l*	1. *ol-std-mn-g*	1. *ol-std-mn-l*
2. *ol-std-md-l*	2. *ol-iqr-mn-l*	1. *ol-std-md-g*	1. *ol-std-md-l*	1. *ol-std-dl-g*
3. *ol-std-mn-g*	3. *ol-std-mn-l*	1. *ol-iqr-mn-g*	1. *ol-std-md-g*	1. *ol-std-dl-l*
3. *ol-std-mn-l*	4. *ol-std-mn-g*	1. *ol-std-md-l*	2. *no-cleaning*	1. *no-cleaning*
4. *ol-iqr-mn-g*	4. *ol-std-md-g*	1. *ol-iqr-mn-l*	2. *ol-iqr-mn-g*	1. *ol-std-md-g*
5. *ol-iqr-mn-l*	4. *ol-std-md-l*	1. *no-cleaning*	2. *ol-iqr-mn-l*	1. *ol-std-md-l*
6. *ol-std-dl-l*	5. *ol-iqr-md-l*	1. *ol-iqr-md-l*	2. *ol-iqr-md-l*	1. *ol-iqr-mn-g*
6. *ol-std-dl-g*	6. *ol-iqr-md-g*	2. *ol-iqr-md-g*	2. *ol-iqr-md-g*	1. *ol-iqr-mn-l*
7. *ol-iqr-md-g*	7. *ol-std-dl-l*	3. *ol-std-dl-l*	3. *ol-std-dl-g*	1. *ol-iqr-dl-l*
7. *ol-iqr-md-l*	7. *ol-std-dl-g*	3. *ol-std-dl-g*	3. *ol-std-dl-l*	1. *ol-iqr-dl-g*
8. *ol-iqr-dl-l*	8. *ol-iqr-dl-g*	4. *ol-iqr-dl-l*	3. *ol-iqr-dl-l*	1. *ol-iqr-md-g*
9. *ol-iqr-dl-g*	8. *ol-iqr-dl-l*	4. *ol-iqr-dl-g*	3. *ol-iqr-dl-g*	1. *ol-iqr-md-l*
1. *le-cl-dl*	1. *le-cl-dl*	1. *le-cl-dl*	1. *le-cl-dl*	1. *le-cl-dl*
2. *le-cl*	2. *le-cl*	2. *le-cl*	2. *le-cl*	1. *le-cl*
3. *no-cleaning*	3. *le-fc*	3. *no-cleaning*	3. *no-cleaning*	1. *no-cleaning*
3. *le-fc*	4. *le-fc-dl*	3. *le-fc*	4. *le-fc*	1. *le-fc-dl*
3. *le-fc-dl*	5. *no-cleaning*	4. *le-fc-dl*	4. *le-fc-dl*	1. *le-fc*
1. *mv-dl*	1. *mv-mn-g*			
2. *mv-mn-g*	1. *mv-mn-l*			
2. *mv-md-g*	1. *mv-md-g*			
2. *mv-mn-l*	2. *mv-cf*			
2. *mv-md-l*	3. *mv-md-l*			
3. *mv-cf*	4. *mv-dl*			

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